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SUPERSEDING
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FEDERAL SPECIFICATION

PIPE, CAST GRAY AND DUCTILE IRON, PRESSURE (FOR WATER AND OTHER LIQUIDS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cast iron (gray iron and ductile iron) pressure pipe for conveying water, sewage, and other liquids.

1.2 Classification.

1.2.1 Types. Pipe shall be of the following types, as specified (see 6.2):

Type I - Calked joint.
Type II - Push-on joint.
Type III - Mechanical joint.

1.2.2 Grades. Pipe shall be of the following grades, as specified (see 6.2):

Grade A - Gray iron 18/40 (18,000 p.s.i. burst strength/40,000 p.s.i. modulus of rupture).
Grade B - Gray iron 21/45 (21,000 p.s.i. burst strength/45,000 p.s.i. modulus of rupture).
Grade C - Ductile iron 60-42-10 (60,000 p.s.i. tensile strength - 42,000 p.s.i. yield strength - 10 percent elongation).

1.2.3 Pressure classes. Pipe shall be of the pressure classes shown in table I, as specified (see 6.2):

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TABLE I. Pressure classes

Pressure class	Liquid working pressure (p.s.i. maximum)
50	50
100	100
150	150
200	200
250	250
300	300
350	350

Gray iron pipe (grades A and B) with a nominal diameter of 30 inches and over shall be classes 50, 100, 150, 200, and 250 only. Ductile iron pipe (grade C), in all sizes, shall be the classes shown in table I.

1.2.4 Sizes. Pipe shall be of nominal diameters from 3 to 58 inches, as specified (see 6.2), in the following increments:

4 to 20 inches - 2-inch increments.
24 to 54 inches - 6-inch increments.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-S-781 - Steel, Strapping, Flat.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Federal Standard:

Fed. Std. 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the

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prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

United States of America Standards Institute (USASI) Publications:

A21.1 - Thickness Design of Cast-Iron Pipe.
A21.4 - Cement Mortar Lining for Cast Iron Pipe and Fittings, Specifications for.
A21.6 - Cast Iron Pipe Centrifugally Cast in Metal Molds, for Water or Other Liquids, Specifications for.
A21.8 - Cast Iron Pipe Centrifugally Cast in Sand-Lined Molds, for Water or Other Liquids, Specifications for.
A21.11 - Rubber Gasket Joints for Cast Iron Pressure Pipe and Fittings.

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A21.50 - Thickness Design of Ductile-Iron Pipe.
A21.51 - Ductile-Iron Pipe. Centrifugally Cast In Metal Molds or Sand-Lined Molds for Water or Other Liquids.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York, 10016.)

Cast Iron Pipe Research Association:

Handbook of Cast Iron Pipe 1967 Edt.

(Application for copies should be addressed to the Cast Iron Pipe Research Association, 3440 Prudential Plaza, Chicago, Ill., 60601.)

American Trucking Associations, Inc.:

National Motor Freight Classification Rules.

(Application for copies should be obtained from the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D. C., 20036.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Ill., 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description.

3.1.1 Type I. Type I pipe shall be furnished either with bell-and-spigot plain end (see figure 1a) or bell-and-spigot beaded end (see figure 1b), and shall be designed for assembly with calked joints. Type I pipe shall be grade A or B made by the centrifugal casting process in metal molds or sand-lined molds. Dimensions, tolerances, and mechanical properties for type I pipe of the specific class required, shall conform to the requirements specified in USASI Std. A21.1, A21.6, and A21.8, as applicable.

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3.1.2 Type II. Type II pipe shall be designed for assembly by prepositioning a continuous, molded, rubber-ring gasket in an annular recess in the pipe socket and forcing the spigot end of the entering pipe into the socket, thereby compressing the gasket radially to the pipe to form a positive seal. The design and shape of the gasket and the annular recess shall be such that the gasket is locked in place against displacement as the joint is assembled. The joint design shall be in accordance with the manufacturer's standard practice. The size and shape of the gasket shall be such as to provide adequate compressive force between the spigot and the socket after assembly to effect a positive seal under all combinations of joint and gasket tolerances. The properties of the rubber gasket shall be as specified in USASI Std. A21.11. The spigot end of the pipe shall be suitably beveled to facilitate assembly. When specified (see 6.2), and prior to the award of the contract, the manufacturer shall furnish the procuring activity drawings of the joint and gasket.

3.1.2.1 Type II pipe shall be centrifugally cast either in a metal mold or a sand mold with use of grade A, B, or C iron. Unless otherwise specified (see 6.2), the specified class of type II pipe shall conform to the dimensions, tolerances and chemical and mechanical properties specified in USASI Std. A21.1, A21.6, A21.8, or A21.51, as applicable.

3.1.3 Type III. Type III pipe shall be furnished with a mechanical joint of the stuffing-box type, having a sealing gasket, a separate follower gland, and tee-head bolts with nuts (see figure 1c). Unless otherwise specified (see 6.2), type III pipe shall be centrifugally cast either in a metal mold or a sand-lined mold with use of grade A, B, or C iron. The specified class of type III pipe, cast from gray iron and ductile iron shall conform to the dimensions, chemical properties, and mechanical properties specified in USASI Std. A21.6, A21.8, or A21.51, as applicable.

3.1.3.1 Type III, sealing gasket and lubricant. Type III sealing gasket and gasket lubricant shall conform to the requirements specified in USASI Std. A21.11.

3.1.3.2 Glands, bolts, and nuts. Requirements for glands, bolts, and nuts shall conform to USASI Std. A21.11.

3.2 Pipe thicknesses. Unless otherwise specified (see 6.2), pipe shall be furnished for installation with 5 feet of cover under laying condition B. For these conditions, thicknesses of pipe shall be as follows: for pipe made of grade A, as given in table 6.4 in USASI Std A21.6 and in table 8.4 of USASI Std. A21.8; for pipe made of grade B,

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as given in table 1-2 in USASI Std. A21.1. Except for 54-inch diameter pipe, which shall be in accordance with the Cast Iron Pipe Research Association Handbook of Cast Iron Pipe, the conditions for grade C pipe shall be in accordance with USASI Std. A21.50 and A21.51.

Laying condition	Description
A	Flat-bottom trench; backfilling not tamped.
B	Flat-bottom trench; backfilling tamped.
F	Pipe bedded in gravel or sand, backfilling tamped.

Pipe thicknesses shall be in accordance with the applicable requirements of USASI Std. A21.1 or A21.50 when any of the following conditions apply:

(a) Pipe to be installed under laying conditions A or F.
(b) Cover to be other than 5 feet for laying condition B.

3.3 Laying length. The total laying length of pipe as installed shall be as specified (see 6.2). Nominal laying lengths of individual pipes shall be 12-, 16-, 18-, or 20-feet, and as given in the applicable USASI Standards, depending on the method of casting. The nominal laying length shall be measured from the shoulder of the socket to the spigot end.

3.3.1 Laying length tolerance. For grades A and B pipe a maximum of 10 percent of the total number of pipe of each size in an order may be furnished as much as 24 inches shorter than the nominal laying length. An additional 10 percent may be furnished as much as 3 inches shorter than the nominal laying length. For grade C pipe, a maximum of 20 percent of the total number of pipe of each size in an order may be furnished as much as 24 inches shorter than the nominal laying length. An additional 10 percent may be furnished as much as 6 inches shorter than the nominal laying length.

3.4 Weight. The weights of pipe shall be as specified in USASI Std. A21.1, A21.6, A21.8, and A21.51, as applicable.

3.5 Coating. Unless otherwise specified (see 6.2), pipe shall be coated inside and outside with a bituminous coating conforming to USASI Std. A21.6, A21.8, and A21.51. When cement-mortar lining is specified (see 3.6) pipe shall be coated on the outside with a bituminous coating.

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3.5.1 Coating on glands. Unless otherwise specified (see 6.2), glands shall be coated with a bituminous coating conforming to USASI A21.11.

3.6 Cement-mortar lining. When specified (see 6.2 and 6.3), pipe shall be lined with cement mortar. The cement-mortar lining shall be in accordance with USASI Std. A21.4, and shall be standard thickness unless otherwise specified (see 6.2).

3.7 Identification marking. Each pipe shall have the weight and thickness class designation painted conspicuously on it. Pipe may be weighed after coating. Thickness class designations shall be as given in table 1-2 of USASI Std. A21.1; as given in table 6.2 of USASI Std. A21.6; as given in table 8.2 of USASI Std. A21.8 and in table 51.1 of USASI Std. A21.51. Each pipe shall be cast or stamped with the manufacturer's mark, and the year in which the pipe was manufactured. If ductile iron is used, the letters DI or "DUCTILE" shall also be cast or stamped on the pipe. All required markings shall be clear and legible. All cast marks shall be on or near the bell.

3.8 Workmanship. The quality of workmanship shall meet the standards prevalent among manufacturers who normally produce cast iron pipe.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for examination and testing. Unless otherwise specified (see 6.2), sampling for examination and testing of pipe, sealing gaskets, glands, tee-head bolt and nuts shall be in accordance with manufacturer's standard practice.

4.3 Examination. Pipe, gaskets, glands, nuts, and bolts shall be examined in accordance with USASI Std. A21.4, A21.6, A21.8, A21.11, and A21.51, as applicable, to verify compliance with the dimensional, coating, cement lining, marking and workmanship requirements.

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4.4 Test. Pipe shall be subjected to the applicable tests listed in table II in accordance with the standard specified.

TABLE II. Test requirements

Requirement	Gray iron pipe (grades A and B)		Ductile-iron pipe (grade C)
	Cast in metal molds	Cast in sand-lined molds	Cast in metal or sand-lined molds
USASI Standard for test methods	A21.6	A21.8	A21.51
Material composition	Chemical analysis	Chemical analysis	Not applicable
Resistance to leakage	Hydrostatic test	Hydrostatic test	Hydrostatic test
Tensile strength	Full length burst	Full length burst	Tensile test
Min. yield strength	Not applicable	Not applicable	Tensile test
Min. elongation	Not applicable	Not applicable	Tensile test
Modulus of rupture	Talbot-strip and ring tests	Talbot-strip and ring tests	Not applicable
Modulus of elasticity	Talbot-strip test	Talbot-strip test	Not applicable
Breaking load and deflection	Not applicable	Test-bar test	Not applicable
Hardness	Rockwell hardness test	Not applicable	Not applicable
Impact test	Not applicable	Not applicable	Notch charpy test

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4.4.1 Rubber gasket test. Rubber gasket material for type II and type III pipe shall be tested as specified in USASI Std. A21.11.

4.4.2 Glands, nuts, and bolts. Glands, nuts, and bolts shall be tested as specified in USASI Std. A21.11.

4.4.3 Cement-mortar lining test. Finished cement-mortar lining shall be tested as specified in USASI Std. A21.5.

4.5 Inspection for preparation of delivery. The packing and marking of pipe and joint material shall be inspected to determine conformance with the applicable requirements of section 5 of this specification. The packing and marking requirements of section 5 covered by referenced specifications and standards shall be inspected in accordance with the applicable requirements of these documents. For levels A and B packing, the packing requirements of section 5 not covered by referenced specifications shall be examined for the defects listed in 4.5.1 in accordance with MIL-STD-105, based on an AQL of 4.0 percent defects.

4.5.1 Classification of defects: Packing.

Categories	Defects
Major	
101	Pipe not bundled as specified.
102	Cushioning and bracing not provided.
103	Strapping not of specified type and class.
104	Boxes not modified by the addition of skids when required.
105	Gross weight of shipping containers exceeds specified limit.
Minor	
201	Not marked as specified.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Pipe. The pipe shall be shipped loose. When specified (see 6.2), pipe 3 inches in diameter, of like kind and size, shall

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be bundled in quantities specified by the procurement agency and strapped with type I, class B flat steel strapping conforming to QQ-S-781. The straps shall be placed 1/6 the length of the bundle from each end with intermediate straps placed equidistant but not exceeding 36 inches on center.

5.1.1.2 Joint material (bolts, nuts, gaskets, glands, and joint lubricant). Joint material shall be packed in wood-cleated plywood boxes in accordance with PPP-B-601 (overseas type), or in nailed wood boxes in accordance with PPP-B-621 (class 2). Contents shall be cushioned and braced to prevent movement. The cleated plywood boxes shall be closed and strapped in accordance with the appendix to the box specification. Flat steel strapping shall be type I, class B. The gross weight of boxes shall not exceed approximately 200 pounds.

5.1.2 Level B.

5.1.2.1 Pipe. The pipe shall be shipped loose.

5.1.2.2 Joint material (bolts, nuts, gaskets, glands, and joint lubricant). Joint material shall be packed in domestic type and class shipping containers conforming to PPP-B-601 or PPP-B-621. Contents shall be blocked and braced to prevent movement. Strapping a type I, class B conforming to QQ-S-781 shall be in accordance with the appendix thereto. The gross weight of boxes shall not exceed approximately 500 pounds. Boxes exceeding 250 pounds gross weight shall be modified by the addition of wood skids in accordance with the applicable box specification.

5.1.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules, or National Motor Freight Classification Rules.

5.2 Marking (see 6.2).

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, pipe shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, pipe shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Types I, II, and III pipe are for use in conveying water, sewage, or other liquids. Types II and III pipe are used to advantage in wet trenches, where flexibility is required or where excessive expansion or contraction may occur.

6.1.1 Cast iron pipe (grades A and B) and ductile iron pipe (grade C) are equally acceptable for general application. However, when unusually severe conditions are encountered, such as impact stresses, shallow burial beneath vehicle crossings, significant differential settlement, including handling, then grade C should be used.

6.1.2 Temperature limitations. Type I pipe should not be used for any service temperatures over 450° F. or for oil over 300° F. For types II and III pipe, the maximum service temperature will be limited by the rubber gaskets. Where an application with liquid temperatures in excess of 180° F. may exist for type II or III pipe, the manufacturer should be consulted.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type, pressure, class, grade, and nominal diameter and thickness of pipe (see 1.2, 3.1.2.1, 3.1.3 and 3.2).

(c) If drawings of gasket for type II pipe is required (see 3.1.2).

(d) If other than 5 feet of cover or if different laying condition is required (see 3.2).

(e) Total laying length (length installed) required (see 3.3).

(f) If coating other than bituminous is required on pipe and glands (see 3.5 and 3.5.1).

(g) Cement-mortar lining if required and thickness if other than standard (see 3.6 and 6.3).

(h) If sampling for examination and testing is different (see 4.2).

(i) Level of packing required (see 5.1).

(j) When pipe 3 inches in nominal diameter shall be bundled and the number of pipe per bundle (see 5.1.1).

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(k) Marking required (see 5.2).

6.3 Lining. When it is known that the pipe will carry water that will cause tuberculation, cement-mortar lining should be specified. When special conditions require, linings can be furnished in thicknesses equal to twice those specified in USASI Std. A21.4.

6.4 Fittings. Fittings for use with pipe covered by this specification are covered by USASI Std. A21.10 entitled "Cast-Iron Fittings, 2 in. through 48 in., for Water and Other Liquids".

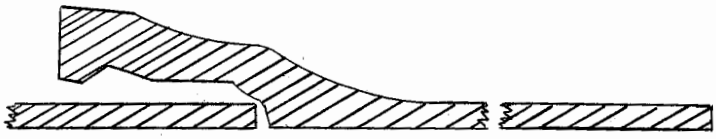
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Custodians:
Army - ME
Navy - YD
Air Force - 11
Review:
Army - ME, CE
Navy - YD
Air Force - 11
User:
Army - MR
Navy - CG, MC

Preparing activity:
Navy - YD

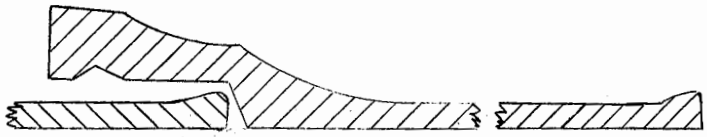
CIVIL AGENCY INTEREST:

GSA - FSS

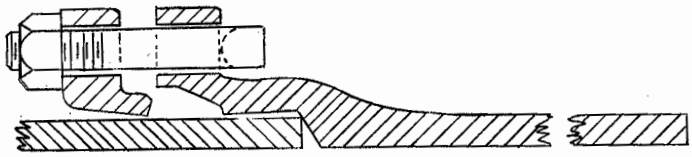
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a. Type I. Pipe with plain end.



b. Type I. Pipe with beaded end.



c. Type III. Pipe.

Note: Dimensions for types I and III pipe shall conform to USA Standards A21.6, A21.8, and A21.51 as applicable.

Figure 1. Types I and III pipe.

Orders for this publication are to be placed with _____, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein.

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